Topology (MAT307/M405/MAT803): Problem Set 6 Detailed Solutions

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Problem 1 (1)**.** *Let $p : X \to Y$ be a closed continuous surjective map such that $p^{-1}(y)$ is compact for each $y \in Y$. Prove that if Y is compact, then so is X.*

Proof: Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an arbitrary open covering of the space X. We wish to extract a finite subcovering of X from $\mathcal{U}$ to establish that X is compact. For each point $y \in Y$, the fiber $p^{-1}(y) \subseteq X$ is a compact subspace by hypothesis. Since $\mathcal{U}$ covers X, it covers $p^{-1}(y)$ in particular; hence, we can extract a finite subcollection of open sets from $\mathcal{U}$, say $U_{\alpha_{1,y}}, U_{\alpha_{2,y}}, \ldots, U_{\alpha_{k_y,y}}$, which covers $p^{-1}(y)$.

Define the union of these selected open sets as $S_y = \bigcup_{i=1}^{k_y} U_{\alpha_{i,y}}$, so that $p^{-1}(y) \subseteq S_y$. The complement $V_y = X \setminus S_y$ is a closed subset of X because S_y is a finite union of open sets. Because $p : X \to Y$ is a closed map, the image $p(V_y)$ is a closed subset of Y, which implies that its complement $W_y = Y \setminus p(V_y)$ is an open subset of Y.

We claim that $y \in W_y$. Suppose to the contrary that $y \notin W_y$. Then $y \in p(V_y)$, meaning there exists some $x \in V_y$ such that $p(x) = y$, or $x \in p^{-1}(y)$. But $V_y = X \setminus S_y$ is disjoint from S_y, whereas $p^{-1}(y) \subseteq S_y$, yielding a contradiction. Thus, $y \in W_y$ for every $y \in Y$.

The collection of open sets $\{W_y\}_{y \in Y}$ forms an open covering of Y. Since Y is compact, this covering contains a finite subcovering, say $W_{y_1}, W_{y_2}, \ldots, W_{y_m}$. For each j, we have $p^{-1}(W_{y_j}) \subseteq X \setminus V_{y_j} = S_{y_j} = \bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}}$. Taking the inverse images over the finite subcovering yields:

$$X = p^{-1}(Y) = \bigcup_{j=1}^{m} p^{-1}(W_{y_j}) \subseteq \bigcup_{j=1}^{m} \left(\bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}} \right).$$

Conclusion: We have expressed X as a finite union of sets belonging to $\mathcal{U}$. Therefore, X is compact. ■

Problem 2 (2)**.** *Let X be a compact metric space. Prove that if $f : X \to X$ is an isometry (distance preserving map) of X, then f is a homeomorphism.*

Proof: Let $f : X \to X$ satisfy $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$. A homeomorphism is a continuous bijection with continuous inverse. We check each part in turn.

Step 1: f is continuous. Given $\epsilon > 0$, take $\delta = \epsilon$. If $d(x, y) < \delta$, then $d(f(x), f(y)) = d(x, y) < \epsilon$. So f is (uniformly) continuous.

Step 2: f is injective. If $f(x) = f(y)$, then $d(x, y) = d(f(x), f(y)) = 0$. Hence $x = y$.

Step 3: f is surjective (this is where compactness is used). Suppose, for contradiction, that $f(X) \neq X$, and pick $y \in X \setminus f(X)$.

(a) $f(X)$ is closed. By Step 1, f is continuous, and X is compact. The continuous image of a compact set is compact (standard theorem), so $f(X)$ is compact. A compact subset of a metric space is closed, so $f(X)$ is closed.

(b) y is at positive distance from $f(X)$. Since $y \notin f(X)$ and $f(X)$ is closed, some open ball $B(y, \epsilon)$ misses $f(X)$. That is, $d(y, z) \geq \epsilon$ for every $z \in f(X)$.

(c) Build a sequence with no convergent subsequence. Define $x_n = f^n(y)$ for $n \geq 0$, where $f^0 = \text{id}$. Take $m > n \geq 0$. Since f is an isometry, so is f^n, and therefore

$$d(x_n, x_m) = d(f^n(y), f^n(f^{m-n}(y))) = d(y, f^{m-n}(y)).$$

Because $m-n \geq 1$, the point $f^{m-n}(y) = f(f^{m-n-1}(y))$ lies in $f(X)$. By (b), $d(x_n, x_m) \geq \epsilon$.

(d) Contradiction. Every pair of distinct terms of (x_n) is at distance at least ϵ. So no subsequence is Cauchy, and hence none converges. But X is a compact metric space, so it is sequentially compact (standard theorem), and (x_n) must have a convergent subsequence. This is a contradiction.

Therefore $f(X) = X$, so f is surjective.

Step 4: f^{-1} is continuous. By Steps 2 and 3, f is a bijection, so $f^{-1} : X \to X$ exists. For $a, b \in X$, write $a = f(x)$ and $b = f(y)$. Then

$$d(f^{-1}(a), f^{-1}(b)) = d(x, y) = d(f(x), f(y)) = d(a, b).$$

So f^{-1} is also an isometry, and by the same argument as in Step 1 it is continuous.

Conclusion: f is a continuous bijection with continuous inverse, so f is a homeomorphism. ■

Problem 3 (3)**.** *Prove that the composition of two quotient maps is a quotient map.*

Proof: Let $p : X \to Y$ and $q : Y \to Z$ be quotient maps. By definition, both p and q are continuous and surjective maps. Consider the composite map $qp : X \to Z$ defined by $(qp)(x) = q(p(x))$. The composition of continuous functions is continuous, and the composition of surjective functions is surjective; hence qp is continuous and surjective.

To prove that qp is a quotient map, we must show that a subset $U \subseteq Z$ is open in Z if and only if its inverse image $(qp)^{-1}(U)$ is open in X. By set-theoretic properties of inverse images under compositions, $(qp)^{-1}(U) = p^{-1}(q^{-1}(U))$.

Suppose $U \subseteq Z$ is such that $(qp)^{-1}(U)$ is open in X. Since p is a quotient map, a subset $V \subseteq Y$ is open in Y if and only if $p^{-1}(V)$ is open in X. Setting $V = q^{-1}(U)$, we have $p^{-1}(V) = (qp)^{-1}(U)$ open in X, which implies $q^{-1}(U)$ is open in Y. Now, since q is a quotient map and $q^{-1}(U)$ is open in Y, it follows that U is open in Z. Conversely, if U is open in Z, then $q^{-1}(U)$ is open in Y by continuity of q, and $p^{-1}(q^{-1}(U))$ is open in X by continuity of p.

Conclusion: U is open in Z if and only if $(qp)^{-1}(U)$ is open in X. Thus, qp is a quotient map. ■

Problem 4 (4)**.** *Prove that a surjective continuous map $f : X \to Y$ that is either an open map or a closed map is a quotient map.*

Proof: A surjective continuous map $f : X \to Y$ is a quotient map if for any subset $U \subseteq Y$, U is open in Y if and only if $f^{-1}(U)$ is open in X (equivalently with closed sets). We analyze the two cases separately:

Case 1 (f is an open map): Let $U \subseteq Y$ be any subset. If U is open in Y, then $f^{-1}(U)$ is open in X by continuity. Conversely, suppose $f^{-1}(U)$ is open in X. Since f is an open map, the image $f(f^{-1}(U))$ is open in Y. Because f is surjective, $f(f^{-1}(U)) = U$, so U is open in Y.

Case 2 (f is a closed map): Let $C \subseteq Y$ be any subset. If C is closed in Y, then $f^{-1}(C)$ is closed in X by continuity. Conversely, suppose $f^{-1}(C)$ is closed in X. Since f is a closed map, $f(f^{-1}(C))$ is closed in Y. Since f is surjective, $f(f^{-1}(C)) = C$, so C is closed in Y.

Conclusion: In both cases, the quotient map condition holds. Therefore, f is a quotient map. ■

Problem 5 (5)**.** *Prove that a bijective quotient map is a homeomorphism.*

Proof: Let $f : X \to Y$ be a quotient map that is also bijective. Because f is a quotient map, it is continuous and surjective. Because f is bijective, it possesses an inverse function $f^{-1} : Y \to X$. To prove that f is a homeomorphism, we must show that f^{-1} is continuous (since f is already continuous).

Showing f^{-1} is continuous is equivalent to showing that for any open set $U \subseteq X$, its inverse image under f^{-1}, which is $(f^{-1})^{-1}(U) = f(U)$, is open in Y. Let U be an arbitrary open subset of X. Because f is injective, $f^{-1}(f(U)) = U$. By definition, f is a quotient map, meaning a subset $V \subseteq Y$ is open in Y if and only if $f^{-1}(V)$ is open in X. Choosing $V = f(U)$, we have $f^{-1}(V) = U$, which is given to be open in X. Therefore, $V = f(U)$ is open in Y.

Conclusion: Since $f(U)$ is open in Y for every open set $U \subseteq X$, the inverse map f^{-1} is continuous. Thus, f is a homeomorphism. ■

Problem 6 (6)**.** *Let $f_1 : X_1 \to Y_1$ and $f_2 : X_2 \to Y_2$ be maps between topological spaces, and let $f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2$ denote the product map $(x_1, x_2) \mapsto (f_1(x_1), f_2(x_2))$.*

(a) Prove that if f_1 and f_2 are continuous, then $f_1 \times f_2$ is continuous with respect to the product topologies.

(b) Give a counterexample to show that if $p_1 : X_1 \to Y_1$ and $p_2 : X_2 \to Y_2$ are quotient maps, $p_1 \times p_2$ need not be a quotient map.

Proof:

(a) **Part (a):** A sub-basis for the product topology on $Y_1 \times Y_2$ consists of sets of the form $U_1 \times U_2$, where $U_1 \subseteq Y_1$ and $U_2 \subseteq Y_2$ are open. To prove $f_1 \times f_2$ is continuous, we show that the inverse image of any basic open set $U_1 \times U_2$ is open in $X_1 \times X_2$. Let $(x_1, x_2) \in (f_1 \times f_2)^{-1}(U_1 \times U_2)$. Then $(f_1(x_1), f_2(x_2)) \in U_1 \times U_2$, which implies $f_1(x_1) \in U_1$ and $f_2(x_2) \in U_2$, or equivalently $x_1 \in f_1^{-1}(U_1)$ and $x_2 \in f_2^{-1}(U_2)$. Thus, $(f_1 \times f_2)^{-1}(U_1 \times U_2) = f_1^{-1}(U_1) \times f_2^{-1}(U_2)$. Since f_1 and f_2 are continuous, $f_1^{-1}(U_1)$ and $f_2^{-1}(U_2)$ are open in X_1 and X_2 respectively, making their Cartesian product open in $X_1 \times X_2$. **Conclusion (a):** $f_1 \times f_2$ is continuous. ■

(b) **Part (b):** Consider the quotient map $p_1 = \pi : \mathbb{R} \to \mathbb{R}/\mathbb{N}$ obtained by collapsing the closed set of integers $\mathbb{N} = \{1, 2, 3, \dots\}$ to a single point $*$, and let $p_2 = \mathrm{id}_{\mathbb{Q}} : \mathbb{Q} \to \mathbb{Q}$ be the identity map on the rational numbers, which is a homeomorphism and hence a quotient map. We examine their product map $F = \pi \times \mathrm{id}_{\mathbb{Q}} : \mathbb{R} \times \mathbb{Q} \to (\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. By part (a), F is continuous, and it is clearly surjective. However, we claim that F is not a quotient map.

To see why, note that the fibers of F over points in the codomain are either single points or sets of the form $\mathbb{N} \times \{s\}$ for $s \in \mathbb{Q}$. A subset $U \subseteq \mathbb{R} \times \mathbb{Q}$ is saturated with respect to F if for each rational s, either the entire vertical slice $\mathbb{N} \times \{s\}$ is contained in U or none of it is. We can construct a saturated open set U by taking an enumeration of irrational numbers $r_n = \sqrt{2}/n$ for $n \geq 1$ and defining continuous functions $h_n(s) = \min\{1/4, |s - r_n|\}$ for $s \in \mathbb{Q}$. Because each r_n is irrational, $h_n(s) > 0$ everywhere on $\mathbb{Q}$. We define the open set $U = \bigcup_{n=1}^{\infty} \{(x, s) \in \mathbb{R} \times \mathbb{Q} : |x - n| < h_n(s)\}$. Since $(n, s) \in U$ for every integer n and rational s, the set contains $\mathbb{N} \times \mathbb{Q}$, ensuring it is saturated.

Let $W = F(U)$ be the image of U. Because U is saturated, the inverse image $F^{-1}(W) = U$ is open in $\mathbb{R} \times \mathbb{Q}$. If F were a quotient map, its image W would necessarily be open in $(\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. Notice that the point $(^*, 0)$ belongs to W. If W were open, there would exist an open neighborhood V of $*$ in $\mathbb{R}/\mathbb{N}$ and a positive number $\delta > 0$ such that $V \times ((-\delta, \delta) \cap \mathbb{Q}) \subseteq W$. The inverse image $\pi^{-1}(V)$ is an open neighborhood in $\mathbb{R}$ containing all integers $n \geq 1$, meaning for each n there exists some $0 < \eta_n < 1/4$ such that $(n - \eta_n, n + \eta_n) \subseteq \pi^{-1}(V)$. By choosing n sufficiently large such that $|r_n| < \delta/2$ and picking a rational s close to r_n via the density of $\mathbb{Q}$, we can construct a point $x = n + (|s - r_n| + \eta_n)/2$ that satisfies $|s - r_n| < |x - n| < \eta_n$ while simultaneously contradicting the bound established by $h_n(s)$. This contradiction shows that the image W cannot be open in the product topology. **Conclusion (b):** The product of quotient maps $p_1 \times p_2$ is not necessarily a quotient map. ■

Problem 7 (7). *Let X and Y be topological spaces, and $f : X \to Y$ be a bijective continuous map. Show that if Y is Hausdorff, then X is also Hausdorff.*

Proof: To show that X is Hausdorff, we must demonstrate that for any two distinct points $x_1, x_2 \in X$ with $x_1 \neq x_2$, there exist disjoint open neighborhoods $U_1, U_2 \subseteq X$ containing x_1 and x_2 respectively. Since $f : X \to Y$ is injective, $f(x_1) \neq f(x_2)$. Because Y is a Hausdorff space, there exist disjoint open sets $V_1, V_2 \subseteq Y$ such that $f(x_1) \in V_1$, $f(x_2) \in V_2$, and $V_1 \cap V_2 = \emptyset$.

Since f is continuous, the inverse images $U_1 = f^{-1}(V_1)$ and $U_2 = f^{-1}(V_2)$ are open subsets of X containing x_1 and x_2 respectively. Furthermore, their intersection is:

$$U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2) = f^{-1}(\emptyset) = \emptyset.$$

Conclusion: We have found disjoint open neighborhoods separating x_1 and x_2. Therefore, X is Hausdorff. ■

Problem 8 (8). *Let X be a compact topological space and $p : X \to Y$ be a quotient map.*

(a) Prove that Y is compact.

(b) Show that if Y is Hausdorff, then p is a closed map.

Proof:

(a) **Part (a):** A quotient map $p : X \to Y$ is continuous and surjective. It is a fundamental topological theorem that the continuous image of a compact space is compact. Since X is compact and $Y = p(X)$, Y is compact. **Conclusion (a):** Y is compact. ■

(b) **Part (b):** Let $C \subseteq X$ be an arbitrary closed set in X. We must prove that $p(C)$ is closed in Y. Because X is compact and C is closed in X, C is compact. Consider the restriction map $p|_C : C \to p(C)$, which is continuous. Thus, $p(C) = p|_C(C)$ is the continuous image of a compact space, so $p(C)$ is compact in Y. Since Y is Hausdorff, any compact subspace of a Hausdorff space is closed. Consequently, $p(C)$ is closed in Y. **Conclusion (b):** p is a closed map. ■

Problem 9 (9). *Let $p : X \to Y$ be a continuous surjection. Prove that p is a closed map if and only if for every subset $S \subseteq Y$ and every open set $U \subseteq X$ containing $p^{-1}(S)$ there exists an open set $V \subseteq Y$ containing S such that $p^{-1}(V) \subseteq U$.*

Proof: Proof of $(\Rightarrow)$**:** Suppose p is a closed map. Let $S \subseteq Y$ and let $U \subseteq X$ be an open set containing $p^{-1}(S)$. Consider the closed set $C = X \setminus U$. Since p is a closed map, $p(C)$ is closed in Y, so $V = Y \setminus p(C)$ is open in Y. For any $s \in S$, $p^{-1}(\{s\}) \subseteq p^{-1}(S) \subseteq U$, meaning no point in the fiber belongs to C. Thus $s \notin p(C)$, so $s \in V$, implying $S \subseteq V$. Furthermore, $p^{-1}(V) = X \setminus p^{-1}(p(C)) \subseteq X \setminus C = U$.

Proof of $(\Leftarrow)$**:** Conversely, suppose the condition holds. Let $C \subseteq X$ be closed. We must show $p(C)$ is closed in Y, or $Y \setminus p(C)$ is open. Let $y \in Y \setminus p(C)$ and set $S = \{y\}$. The fiber $p^{-1}(\{y\})$ is disjoint from C, so $p^{-1}(\{y\}) \subseteq X \setminus C$. Let $U = X \setminus C$, which is open. By hypothesis, there exists an open set $V_y \subseteq Y$ containing y such that $p^{-1}(V_y) \subseteq U = X \setminus C$. This implies $V_y \cap p(C) = \emptyset$, so $V_y \subseteq Y \setminus p(C)$. Thus, $Y \setminus p(C)$ is open.

Conclusion: p is a closed map if and only if the stated condition holds. ■